

Image Derivatives and Edge Detections

1. Derivatives using loops:

a. Forward derivatives:

```
S = im2double(rgb2gray(imread('coins.jpg')));  
[row, col] = size(S);  
R = zeros(row, col);  
    for i = 1:row  
        for j = 1:col-1  
            R(i,j) = S(i, j) - S(i, j+1);  
        end  
    end
```

b. Backward:

```
R(i,j) = S(i, j-1) - S(i, j);
```

c. Central:

```
R(i,j) = S(i, j-1) - S(i, j+1);
```

2. Derivatives using convolution: (Horizontal Derivative)

a. Forward derivatives:

```
h = [1 -1];
```

b. Backward derivatives:

```
h = [-1 1];
```

c. Central derivatives:

```
h = [1 0 -1];
```

3. Prewitt Edge Detection:

```
Gx = [-1 0 1;  
      -1 0 1;  
      -1 0 1];  
Gy = [ 1 1 1;  
      0 0 0;  
      -1 -1 -1];
```

Take the input image I .

Apply Gx and Gy filters to determine derivatives in 2 directions: $\frac{d}{dx}I$ and $\frac{d}{dy}I$.

Calculate magnitude $d(i,j)$ of from the derivatives for each pixel (i,j) using –

$$d = \sqrt{\frac{d}{dx}I^2 + \frac{d}{dy}I^2}$$

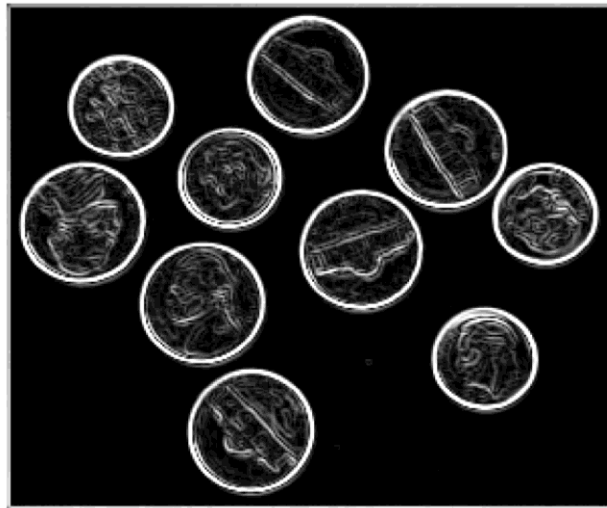
input



edges along x



edges along y



4. Sobel Edge Detection:

$$\begin{aligned} G_x &= \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}; \\ G_y &= \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}; \end{aligned}$$

This mask works exactly the same as the Prewitt operator mask. There is only one difference that is it has “2” and “-2” values in the center of the first and third rows and columns. When applied to an image this mask will highlight the edges.

Steps:

- i. Choose a threshold T .
- ii. Take the input image I .
- iii. Apply G_x and G_y filters to determine derivatives in 2 directions: $\frac{d}{dx}I$ and $\frac{d}{dy}I$.
- iv. Calculate magnitude $d(i, j)$ of from the derivatives for each pixel (i, j) using –

$$d = \sqrt{\frac{d}{dx}I^2 + \frac{d}{dy}I^2}$$

- v. If $d(i, j) > T$ then there is an edge at (i, j) .
- vi. Else, no edge at (i, j) .